

HARISH KUMAR SAINI

Python Backend & Generative AI Engineer | FastAPI | LangChain | LangGraph
RAG | AI Agents | AWS | SQL

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CAREER SUMMARY

Results-driven Python Developer with nearly 4 years of experience at Tata Consultancy Services (TCS), specializing in backend development and workflow automation. Proven expertise in architecting automation solutions using Flask, SQL, and QGIS that reduced manual efforts by up to 80% for international clients in the Telecom. Adept at streamlining complex ETL processes and optimizing database performance to drive operational excellence. Recently focused on building AI-powered applications using LangChain and FastAPI, including RAG-based systems.

TECHNICAL SKILLS

- Programming: Python, SQL
- Agentic AI & Generative AI: LangChain, LangGraph, Multi-Agent Systems, Agent Orchestration, Agent Memory, AI Agent and Agentic Workflow
- LLMs & AI Platforms: OpenAI, Gemini, Hugging Face
- RAG & Prompt Engineering: Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Semantic Search, Vector Search, Function Calling, Structured Outputs, Knowledge Retrieval Systems
- Backend Development: FastAPI, Flask, Rest APIs, API Development, JSON, API Integration, Streaming APIs
- Vector Databases: FAISS, Pinecone, ChromaDB, Weaviate
- Databases: MySQL, PostgreSQL, SQLite
- Data Processing: Pandas, NumPy, ETL Pipelines
- Observability: Langsmith, Logging, Monitoring, Tracing, Audit Logging
- DevOps & Cloud: Git, Docker, AWS (EC2), CI/CD
- Frontend: Streamlit, HTML, CSS
- Tools and Platforms: QGIS, Spirent iTest, Service Now, Jira, PyCharm, Jupyter Notebook, VS Code, IntelliJ IDE, GitHub and Networking

EXPERIENCE

Developer – Tata Consultancy Services (TCS)

Noida | Oct 2021- Aug 2025

- Developed and maintained **Python-based backend automation solutions** using **Flask, SQL, and QGIS**, streamlining complex business workflows and improving data integration across enterprise telecom projects.
- Designed and optimized **ETL pipelines, SQL queries, and workflow automation**, improving processing efficiency, data accuracy, and overall system reliability for multiple client engagements.
- Built scalable automation tools that reduced manual effort by **70–80%**, accelerated delivery timelines, and enhanced operational productivity.
- Collaborated with cross-functional teams to analyze business requirements, troubleshoot production issues, and deliver high-quality automation solutions within project timelines.
- Authored comprehensive technical documentation, conducted knowledge transfer sessions, and provided ongoing production support to ensure seamless solution adoption and maintainability.

PROFESSIONAL PROJECTS

WesBank SARB – BI Portal Support & Optimization (TCS, Belgium & UK)

- Spearheaded the support and enhancement of the BI Portal, ensuring seamless enterprise-level data integration for reporting.
- Engineered and optimized complex SQL scripts for ETL and reporting workflows, significantly improving system uptime by 30% and reducing failures.
- Delivered rapid, high-accuracy resolutions to database queries via ServiceNow, boosting operational efficiency and stakeholder satisfaction.

Network Solutions and Services Delivery - Telecom Workflow & GIS Automation (TCS, Belgium & UK)

- Engineered automation of telecom network design workflows using Python with QGIS and Excel integration, streamlining MACD operations, microwave planning, and cable route identification to achieve a 70% reduction in manual workload.
- Developed a custom QGIS-based Python library to process OSM datasets, identify geographical points, compute distances, and determine optimal routing paths, eliminating manual mapping inefficiencies.
- Delivered scalable automation tools that enhanced documentation accuracy, improved spatial analysis efficiency, and accelerated overall project delivery.

Proximus – Network Operations Automation (TCS, Belgium & UK)

- Automated critical networking operations using Python and Spirent iTest, minimizing manual configurations and accelerating deployment cycles.
- Designed and executed advanced automation scripts for diverse network tasks, significantly improving accuracy, efficiency, and reliability.
- Authored comprehensive technical documentation and conducted knowledge transfer sessions, ensuring seamless adoption and long-term sustainability of automation solutions.

FEATURED AI PROJECTS

1. AI Blog Writing Agent

Tech Stack: LangGraph | LangChain | FastAPI | Streamlit | OpenAI | Tavily | RAG | Docker | AWS EC2

Description:

- Designed and developed a multi-agent AI blog generation system using LangGraph with structured planner, researcher, writer, and reviewer agents to automate end-to-end content creation.
- Integrated Tavily Search API to enable real-time web research for fact-aware and up-to-date blog generation.
- Implemented Retrieval-Augmented Generation (RAG) using vector embeddings (FAISS) for context-aware and domain-specific responses.
- Built and exposed RESTful APIs using FastAPI to orchestrate multi-agent workflows and manage inter-agent communication.
- Enabled streaming-based response generation for real-time content rendering and improved user experience.
- Integrated Generative AI image generation (OpenAI / Gemini) to automatically create featured blog visuals.
- Developed an interactive Streamlit-based frontend for seamless blog creation, monitoring, and user interaction.
- Deployed system using Docker and AWS EC2 for scalable and production-ready deployment.
- GitHub: <https://github.com/Harishsaini1812/AI-Blog-Writing-Agent>

2. RAG Document Question Answering System

Tech Stack: Python | LangChain | FastAPI | FAISS | OpenAI | Streamlit | Docker

Description:

- Built an AI-powered document question answering system using Retrieval-Augmented Generation (RAG).
- Processed PDF documents into semantic chunks and generated embeddings for efficient retrieval.
- Stored embeddings in FAISS vector database for low-latency semantic search.
- Implemented citation-aware responses by retrieving relevant document sections before LLM inference.
- Developed FastAPI endpoints for document upload, indexing, and question answering.
- Created a Streamlit interface supporting PDF upload and interactive Q&A.
- Dockerized the application for consistent deployment across environments.
- GitHub: <https://github.com/Harishsaini1812/rag-document-qa-system>

3. AI Customer Support Chatbot

Tech Stack: Python | LangChain | FastAPI | FAISS | OpenAI | Streamlit

Description:

- Developed an AI-powered customer support chatbot using LangChain and Retrieval-Augmented Generation.
- Built a knowledge base from company documents and FAQs using semantic embeddings.
- Implemented conversation memory to provide context-aware multi-turn interactions.
- Integrated ChromaDB for vector storage and efficient document retrieval.
- Exposed chatbot functionality through FastAPI REST APIs.
- Designed a Streamlit interface for real-time conversations and knowledge search.
- Improved answer accuracy using prompt engineering and retrieval optimization.
- GitHub: <https://github.com/Harishsaini1812/AI-SmartShop-Customer-Support-Chatbot>

4. AI Resume Analyzer & Interview Assistant

Tech Stack: Python | LangChain | FastAPI | OpenAI | Streamlit

Description:

- Developed an AI-powered resume analyzer to evaluate resumes against job descriptions.
- Implemented ATS score estimation and identified missing technical skills.
- Generated personalized interview questions based on resume content and target role.
- Used prompt engineering to produce structured improvement suggestions.
- Developed REST APIs using FastAPI and an interactive Streamlit frontend.
- Supported multiple resume formats including PDF and DOCX.
- GitHub: <https://github.com/Harishsaini1812/AI-Resume-Analyzer-and-Interview-Assistant>

EDUCATION

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| • KCC Institute of Technology and Management, Greater Noida
B. Tech in Information Technology | 2017 – 2021
CGPA: 8.6/10 |
| • Sarvodaya Bal Vidyalaya, Naraina, New Delhi
12 th (PCMB) | 2016 – 2017
Percentage: 65.6% |
| • Sarvodaya Bal Vidyalaya, Naraina, New Delhi
10 th | 2013 – 2014
Percentage: 84.2% |

PERSONAL DETAILS

Date of Birth: 18th December, 1997.
Address: CB-89 Naraina Ring Road, New Delhi-110028.
Marital Status: Unmarried
Languages Known: English and Hindi.
Location Preference: Delhi NCR, Gurugram, Noida.